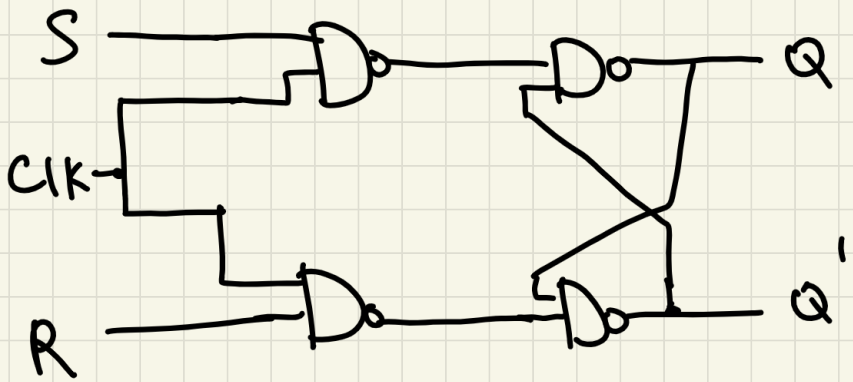
LAB 6

Q1.

a) SR flip flop



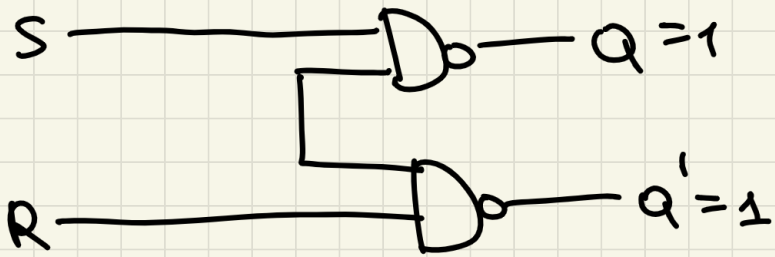
b) Truth table

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

NAND

S	R	Q_{n+1}	State
0	0	Q_n	hold
0	1	0	reset
1	0	1	set
1	1	x	invalid

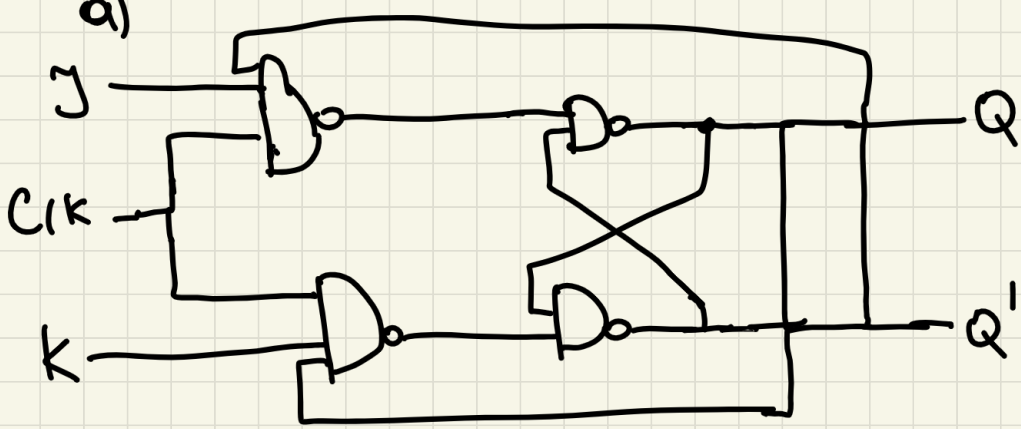
c) Truth table state is when $R=1$ $S=1$, so that NAND gate inputs become 1. That makes $Q=Q'=1$ which is impossible because Q and Q' should be compliments

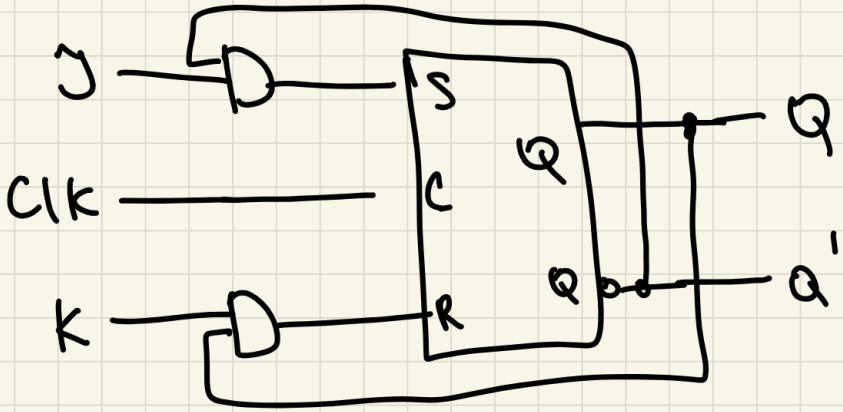


unstable

Q2

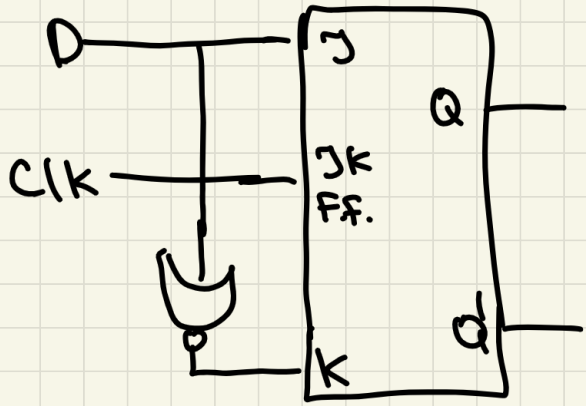
a)





Clk	J	K	Q next	
↑	0	0	Q prev	no change
↑	0	1	0	reset
↑	1	0	1	set
↑	1	1	Q prev	toggle

b)



D flip flop can be implemented using JK flipflop

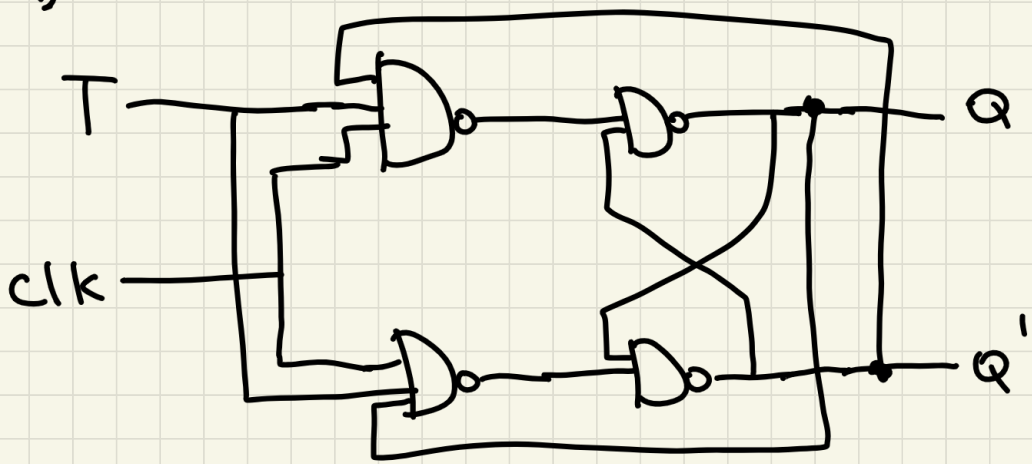
$J \to D$, $K \to D'$.

when D=1, JK sets Q=1,
when D=0, It resets Q=0,
as a standart flipflop.

D	J	K	Q_{next}	
0	0	1	0	reset
1	1	0	1	set

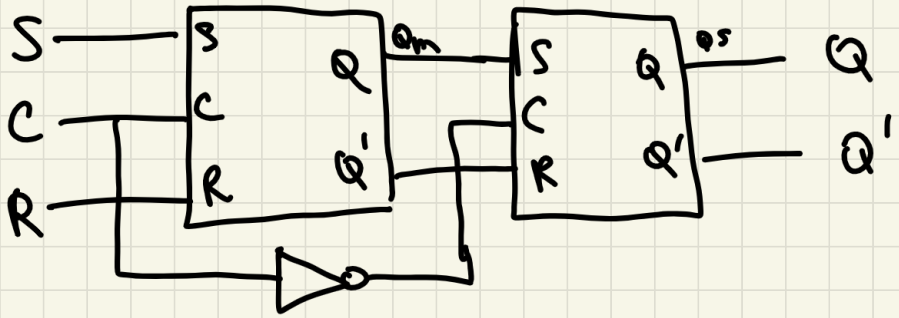
Q3

a)



T	Clk	Q	Oper.
0	↑	Q	Hold
1	↑	Q'	toggle

b)



Sequential enabling in cascaded SR circuits works like a master-slave SR. Two latches are connected in series and controlled by different clock phases. When C=1, the master latch in enabled and reads the inputs, while slave latch is disabled. When C=0, the master latch is disabled and stores its value, while the slave latch becomes enabled and updates final output.

Q4.

a)

Q	Q_{next}	D
0	0	0
0	1	1
1	0	0
1	1	1

$D = Q_{next}$

Q	Q_{next}	T
0	0	0
0	1	1
1	0	1
1	1	0

$T = Q \oplus Q_{next}$

Pres. Q	Q_{next}	D	T	
0	0	0	0	hold at 0
0	1	1	1	set
1	0	0	1	reset
1	1	1	0	hold at 1

b)

Yes, a latch can be a sub-component of a flip-flop. Most flip-flops are built using two latches connected in a master-slave configuration. The master latch captures the input during one clock level, while the slave latch updates the output during the opposite clock level. This arragment allows the flip-flop to operate as a edge-triggered device.

Q5

D latch

D Flip-Flop

E	D	Qnext	State
0	X	Q	hold
1	0	0	reset
1	1	1	set

Truth table D latch

Clk	D	Qnext
↑	0	0
↑	1	1
no edge	X	Q

D flip flop

Differences:

	triggered	controled	complexity	speed
Latch	level	by enable	simple	fast
Flip flop	edge	by clock	complex	stable